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A Place to Learn, A Chance to Grow

Module 4:

Human Population & Environment

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EVS- An Insight

Key Environmental Problems and Sustainability

Global environmental issues, socio-environmental interaction; impact of consumerism and global economy; Sustainability: Fundamentals, science behind sustainability, challenges, potential; role of technology, policies.

Ecosystem and Biodiversity

Ecosystem: functions and services, Nutrient, phosphorous, carbon, nitrogen, cycles; Effect of human activities on these cycles. Threats to biodiversity: conservation strategies, sustainable solutions.

Environmental pollution

Pollution sources, effects and mitigation, conservation. Water footprint and virtual water. Sustainable solutions. Environmental Protection Act – Air, water, forest and wild life. Interactional protocols and agreements. Resource recycling and sustainability.

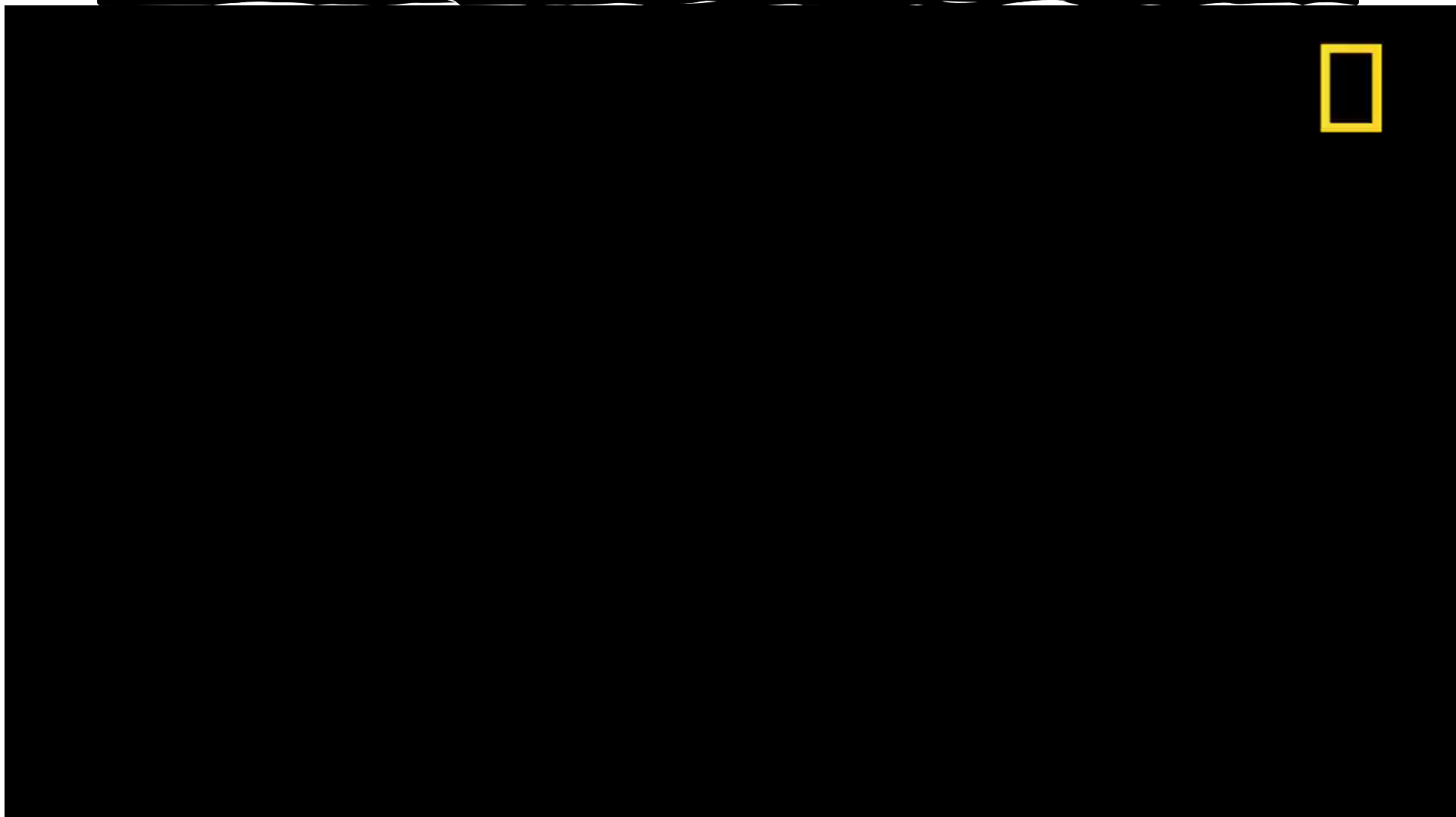
Human Population and Environment

Impact of population age structure, human population growth, Sustaining human societies: Economics, environment, policies and education.

Energy Sources and Sustainability

Conventional sources, non-conventional sources, natural resources and economics, clean coal technology, sequestration methods, energy conservation and sustainability, policies, acts, examples.

7 Billion: 14 Years Ago!!!



Human Population and the environment

Population

It is defined as a group of individuals of the same species living in the same area in a given time.

Population density

It is expressed as the no. of individuals of the population per unit area.

7 Billion: Who is most typical?



Parameters affecting population size

1. **Birth rate or Natality** : It is the no. of birth per 1000 people in a population in a given year.

2. **Death rate**: It is the no. of deaths per 1000 people in a population in a given year.

3. **Immigration** : It denotes the arrival of individual from neighboring population.

4. **Emigration:** It denotes the dispersal of individuals from the original population to new areas.

Population Growth

- ✓ The rapid growth of the global population for the past 100 years results from the difference between the rate of birth and death.

- ✓In 1800 the global population was about 1 billion people. It took 130 years (1930) to reach 2 billion, but the population reached to 4 billion within 45 years (1975).

- ✓Now, we have already crossed 7 Billion and may reach 10 billion by 2050 as per the World Bank calculations.

Causes of Rapid population growth

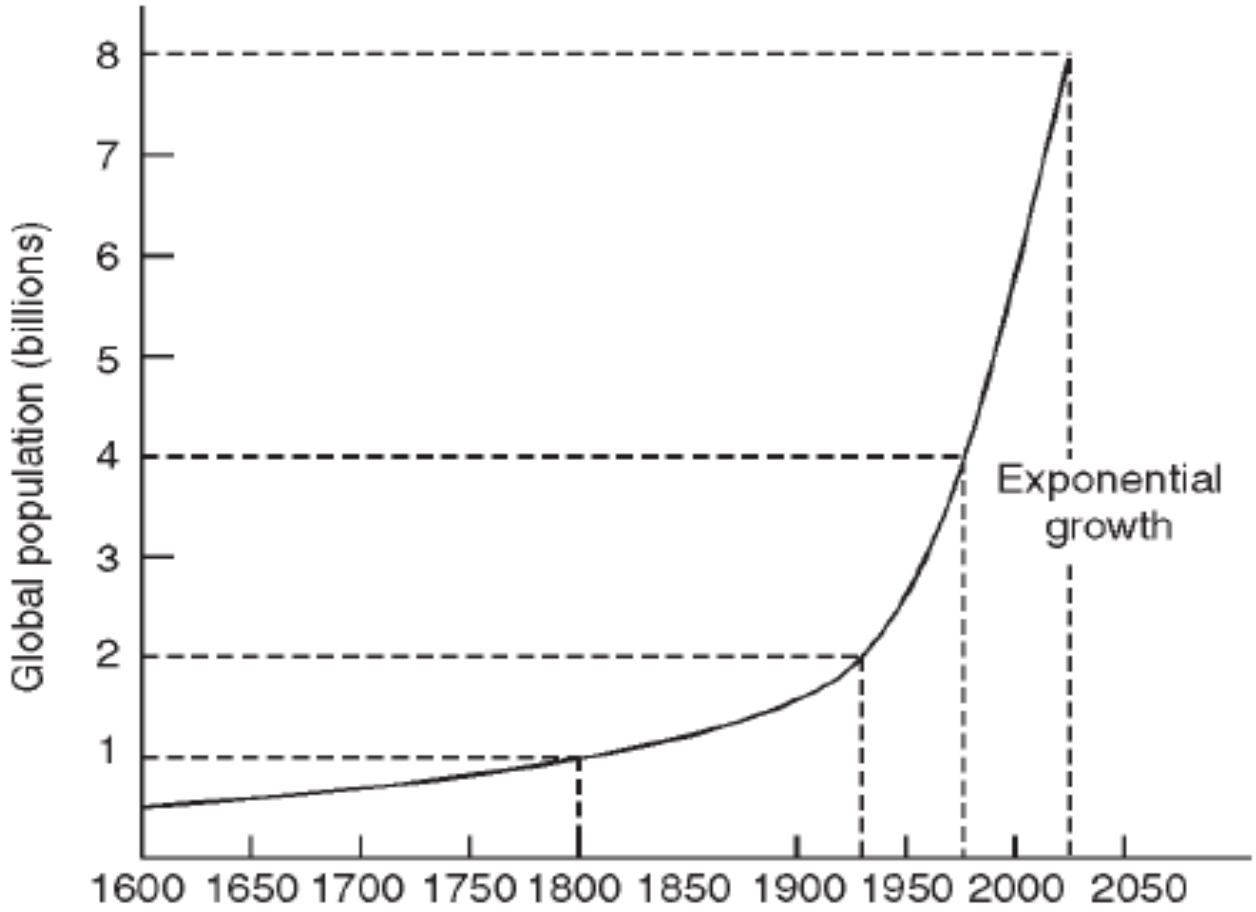
1. It is due to the decrease in death rate and increase in birth rate

2. Availability of antibiotics, immunization, increased food production, clean water, air etc decreases the famine related deaths and infant mortality

3. Better technology

4. ????????

Global population growth trends



Global population growth trends in the last four centuries.

Characteristics of Population growth

1. **Exponential growth:** Now population growth occurs exponentially like 10, 10^2, 10^3, 10^4 etc, which shows the dramatic increase in global population in the past 100 years

2. **Doubling time:** It is the time required for a population to double its size at a constant annual rate. It is calculated as follows

   T_d (doubling time) = 70/r

   Where, r = annual growth rate

   If a nation has 2% annual growth rate, its population will double in next 35 years.

3. Total Fertility rates (TFR)

- It is one of the key measures of a nation's population growth
- TFR is defined as the average number of children that would be born to a woman in her lifetime if the age specific birth rates remain constant
- The value of TFR varies from 1.9 in developed nations to 4.7 in developing nations

4. Infant mortality rate

- It is an important parameter affecting future growth of a population
- It is the percentage of infants died out of those born in a year
- Although this rate has declined in the last 50 years, but the pattern differs widely in developed and developing countries

More factors!!

5. Male - female ratio:

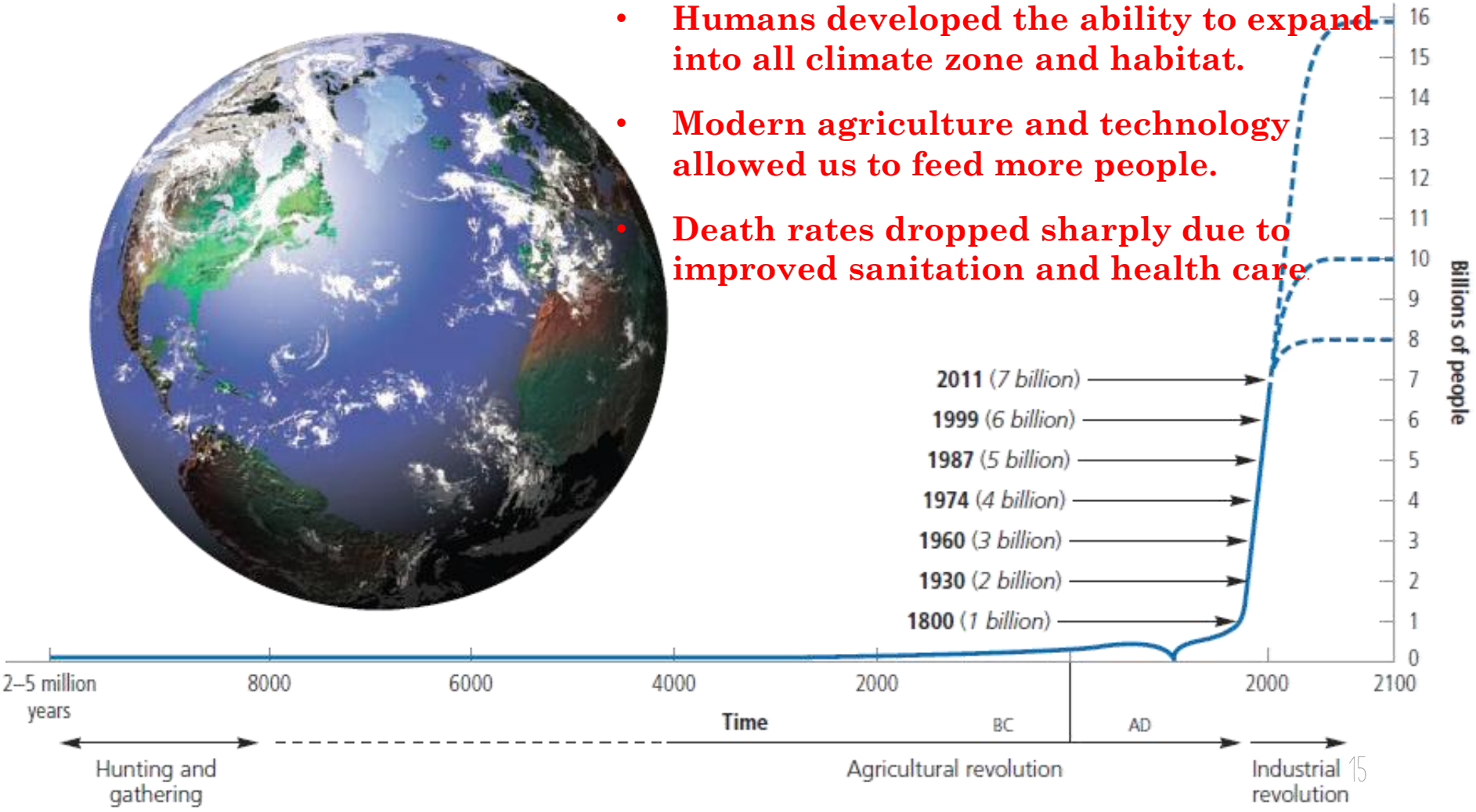
The ratio of girls and boys should be fairly balanced in a society to flourish. But the ratio has been upset in many countries including China and India. In China the ratio of girls and boys is 100:140 but now it has changed for good in India.

6. Demographic transition:

Population growth is generally related to be economic development. The death rates fall due to improved living conditions. This results in the population growth. This phenomenon is referred as demographic transitions.

Global Population Growth

- **Humans developed the ability to expand into all climate zone and habitat.**
- **Modern agriculture and technology allowed us to feed more people.**
- **Death rates dropped sharply due to improved sanitation and health care**



Are there limits to population growth?

Limiting factors are resources or other factors **in** the environment that can lower the **population growth** rate. Limiting factors include a low food supply and lack of space.

The logistic population growth occurs when **the growth** rate of decreases as **the population reaches the** carrying capacity. **The** carrying capacity is **the maximum** number of individuals in a **population** that **the** environment can support.

Reducing biodiversity

Increasing use of net primary productivity

Increasing genetic resistance in pest species and disease-causing bacteria

Eliminating many natural predators

Introducing harmful species into natural communities

Using some renewable resources faster than they can be replenished

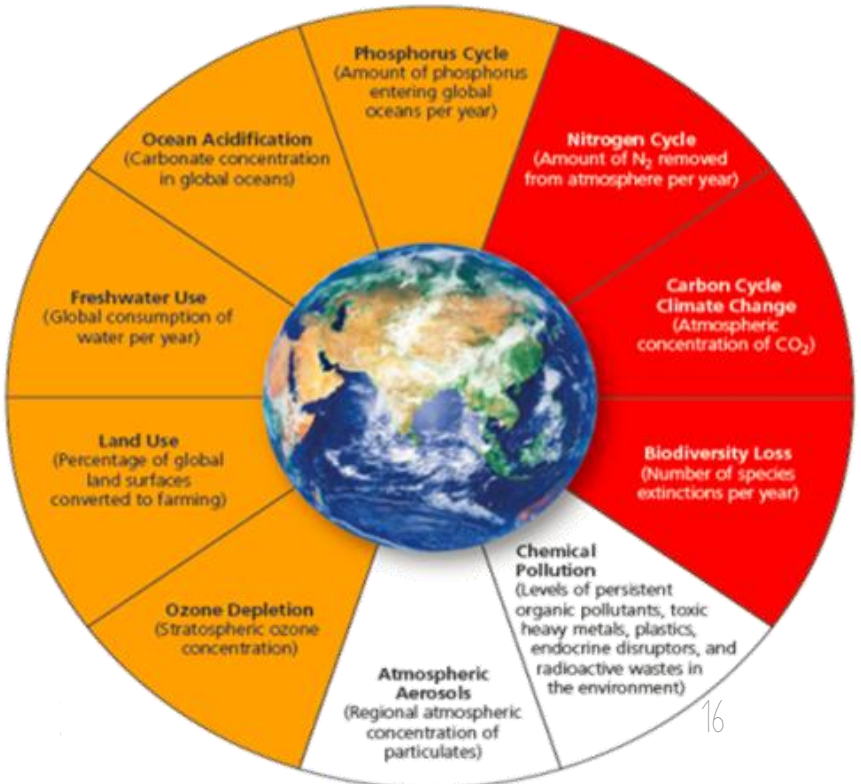
Disrupting natural chemical cycling and energy flow

Relying mostly on polluting and climate-changing fossil fuels







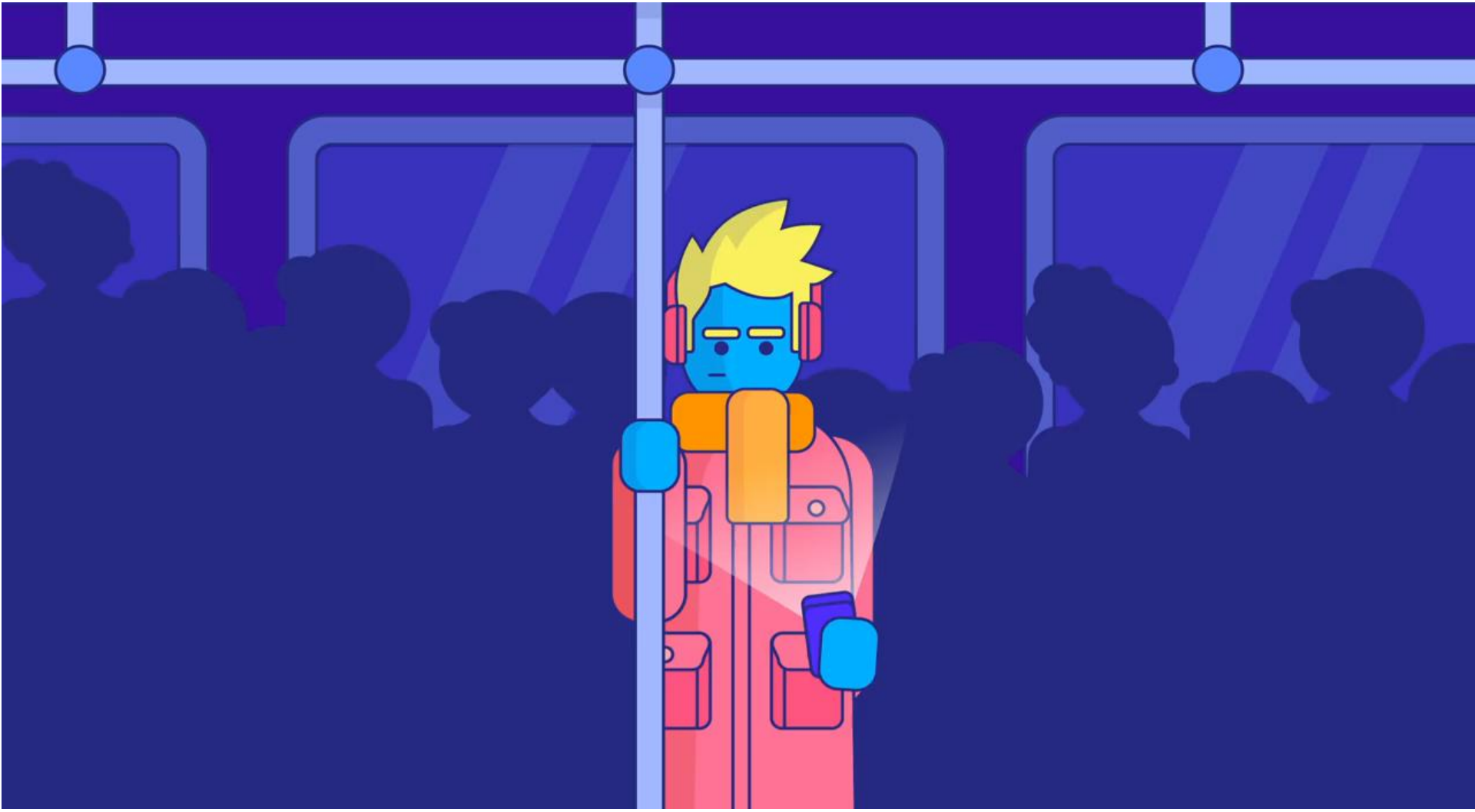


Population Explosion

The enormous increase in population due to low death rate and high birth rate is termed as population explosion. The human population is not increasing at a uniform rate in all parts of the world.

Less Developed Country		Developed country	
Country	Doubling Time	Country	Doubling Time
India	28 years	USA	87 years
Turkey	28	UK	231
Nigeria	27	Italy	99
Saudi Arabia	25	France	117
Pakistan	21	Japan	58

Overpopulation: The Human Explosion!



Causes of population explosion

1. Inventions of modern medical facilities reduces death rate and increases birth rate.

2. Increase of life expectancy is another important reason

Eg. In 1959's life expectancy – 40 years now it is 61 years

3. Illiteracy is another one reason for population explosion.

Effect of population explosion

1. Poverty: Infant mortality is one of the most tragic indicators of poverty. There are still 34 developing countries where more than 1 in 10 children die before reaching the age of five.
2. Population explosion leads to environmental degradation
3. Population explosion causes over exploitation of natural resources.
4. Increase in population increases diseases
5. Over crowding of cities lead to development of slums
6. Unemployment
7. Lack of basic amenities like water supply and sanitation, etc

Problems of Population growth

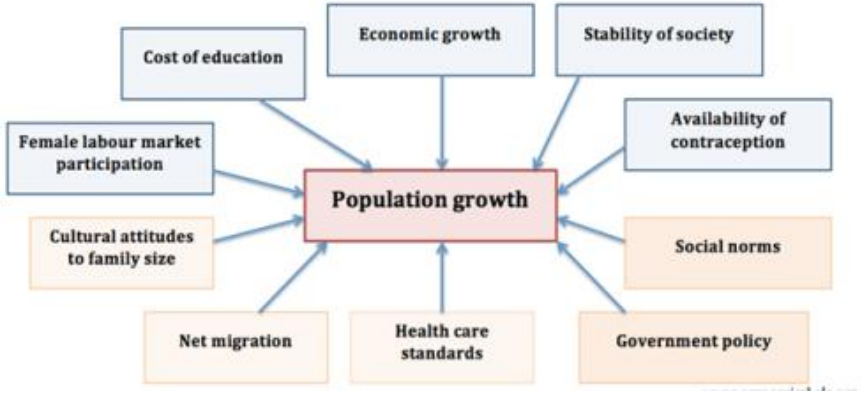
1. Increasing demands for food and natural resources
2. Inadequate housing and health service
3. Loss of agricultural land
4. Unemployment and socio-political unrest
5. Environmental pollution
6. ??

Global Population growth and Impact on Environment

Country	Population Size	Population Growth Rate	Resource Use Per Person	Use of Harmful Technology	Use of Beneficial Technology	Overall Environmental Impact
High-Income Countries						
United States	316 million	Moderate (0.5%)	Very high	Moderate	High	High
Japan	128 million	Negative (−0.2%)	High	Moderate	High	Moderate
Germany	82 million	Negative (−0.2%)	High	Moderate	High	Moderate
Middle-Income Countries						
China	1.35 billion	Moderate (0.5%)	Low	High	Moderate	High
India	1.26 billion	High (1.5%)	Low	High	Low	High
Brazil	194 million	Moderate (1%)	Low	High	Moderate	Moderate
Low-Income Countries						
Nigeria	402 million	High (2.6%)	Very low	High	Low	Moderate
Bangladesh	228 million	High (1.6%)	Very low	High	Low	Moderate
Congo	194 million	High (2.8%)	Very low	High	Low	Moderate

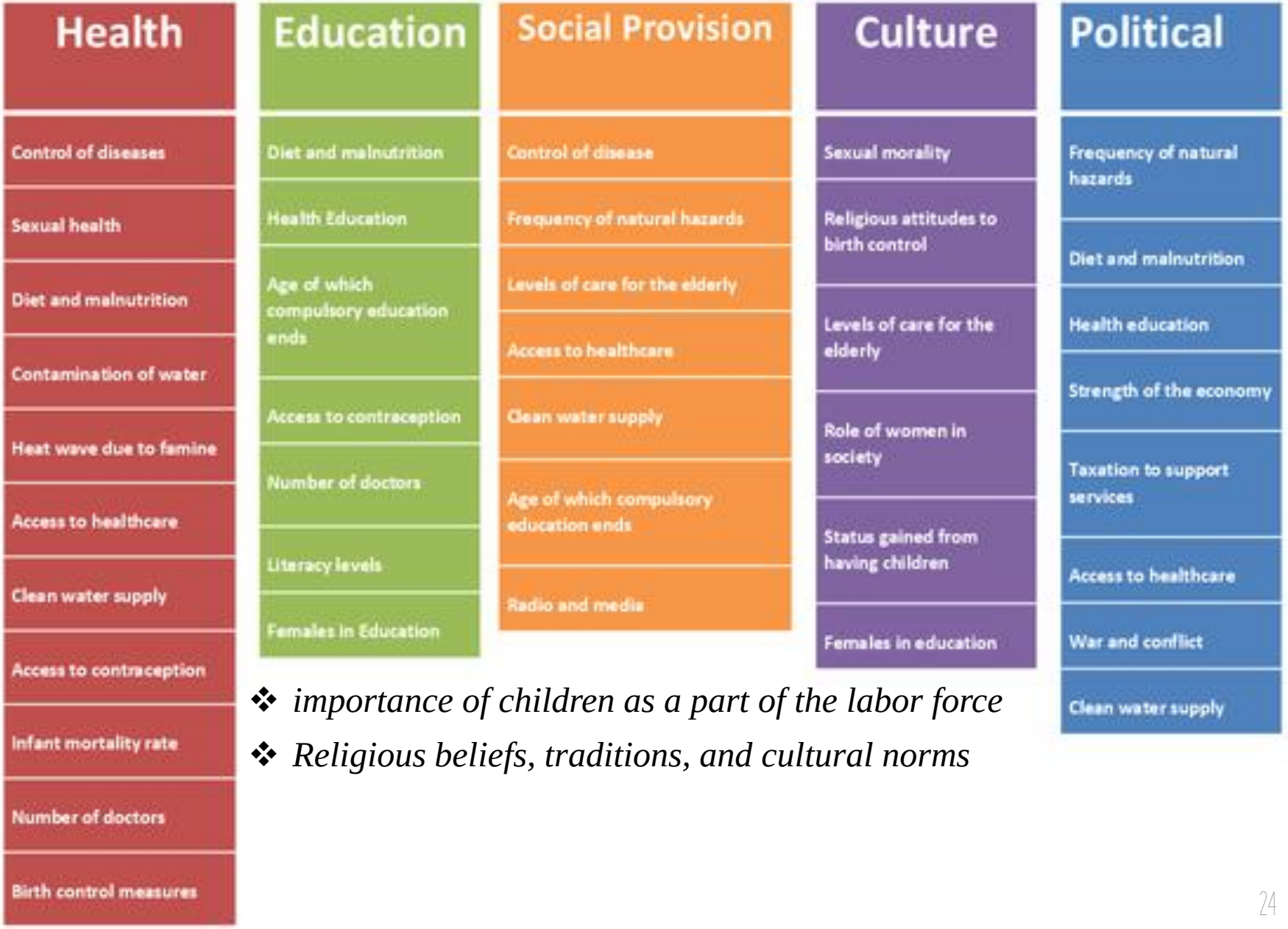
Factors affecting population growth

- Population size increases through births and immigration.
- Population size decreases through deaths and emigration.
- Doubling time : time taken for population to double its size at a constant annual rate
- Total Fertility rate: average no. of children that would be born to a women in her lifetime. **TFR is 1.9 in developed nations and 4.7 developing nations, 1950 it was 6.1**
- Infant mortality rate: % of infants that died out of those born in a year
- Replacement level



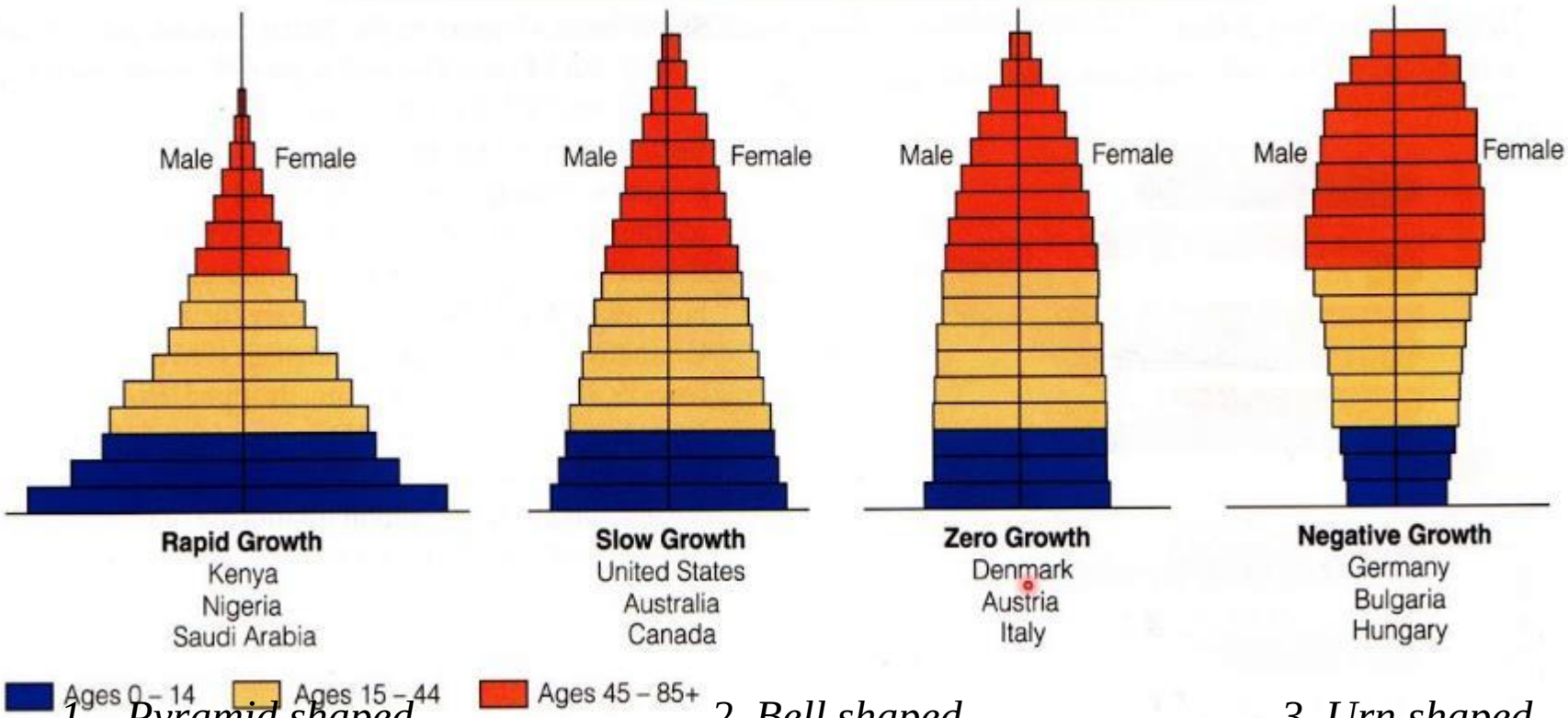
$$\text{Population change} = (\text{Births} + \text{Immigration}) - (\text{Deaths} + \text{Emigration})$$

Factors Influencing Population Growth



- *importance of children as a part of the labor force*
- *Religious beliefs, traditions, and cultural norms*

Types Of Population Pyramids



1. *Pyramid shaped*

Expansive



2. Bell shaped

Stationary



3. Urn shaped

Constrictive



Ways to control population growth

- EDUCATION : **reduce gender inequality**
- AGE OF MARRIAGE SHOULD BE RAISED
- FAMILY PLANNING FACILITIES
- WOMEN-EMPOWERMENT: **reduce gender inequality**
- EASY & CHEAP AVAILABILITY OF CONTRACEPTIVES
- LEGISLATION



- **Improving health care: reduce infant deaths**
- **Legal abortions**
- **Lack of awareness : health, nutrition**
- **Reduce poverty: child labor, human trafficking**



Different methods of contraception: caps, combined pill, condoms (female), condoms (male), contraceptive implant, contraceptive injection, contraceptive patch, diaphragms, intrauterine device (IUD), intrauterine system (IUS), natural family planning, progestogen-only pill, vaginal ring

There are 2 permanent methods of contraception:

- female sterilisation
- male sterilisation (vasectomy)

Population control via economic development

Developed Nations (First World Nations)	*vs.*	*Developing Nations (Third World Nations)*
Plenty of food		Not enough food
Quality health care		Poor health care
High life expectancy		Lower life expectancy
Better jobs / higher pay		Low income jobs
High standard of living		Low standard of livin
Higher literacy rate		Low literacy rate
Better education		Many uneducated
Commercial farming		Subsistence Farming

- Researchers claim that the solution to growth is industrialization and economic development.
- in developed countries:
  - per capita incomes rise,
  - poverty declines,
  - populations tend to grow more slowly.
  - people tend to choose a better standard of living instead of having a large family.



Other Aspect:

improved standard of living may lead to **increase the carrying capacity of the Earth** and lack of water and food !!!!!!!

Population control via Empowerment of Women

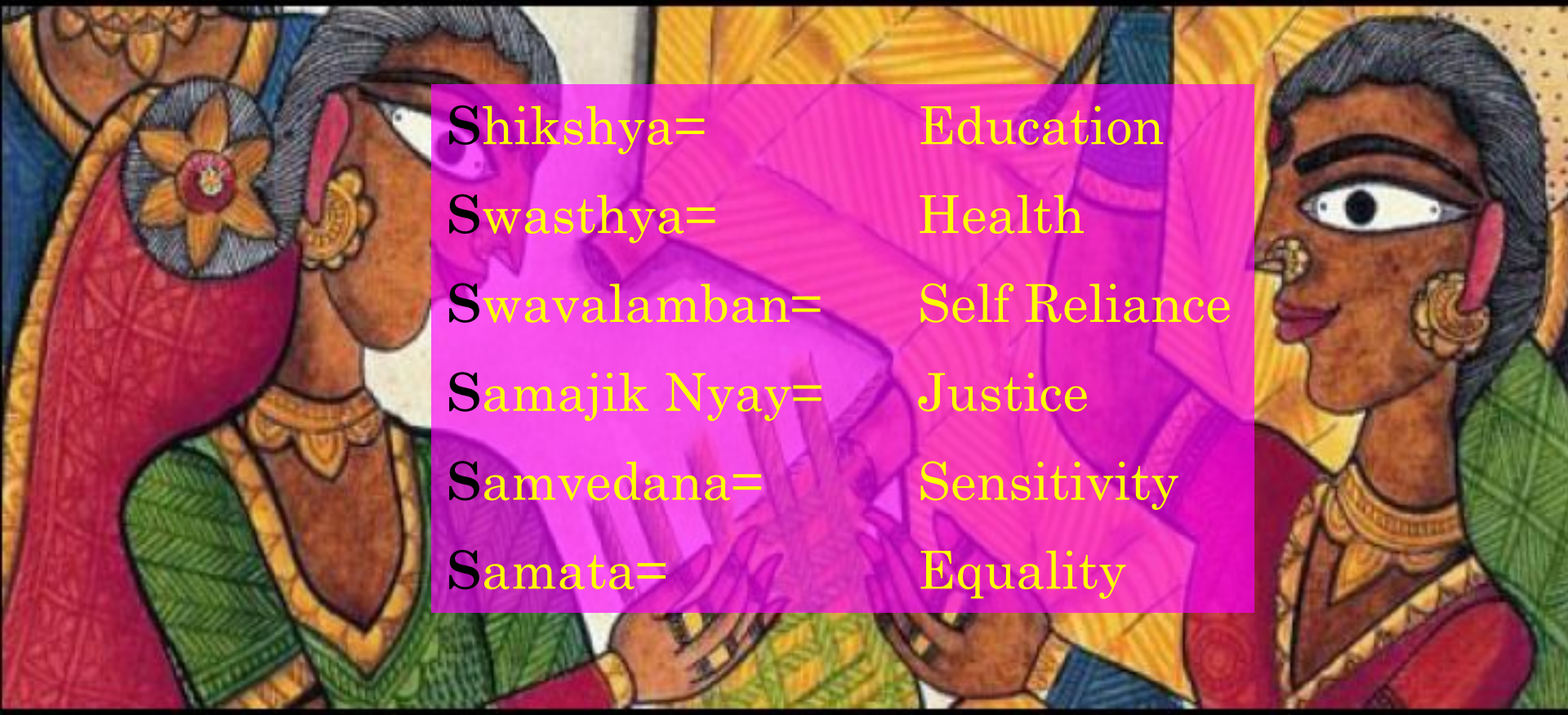
- A number of studies show that underdeveloped **women who cannot read often have an average of five to seven children**
- A growing number of women in less-developed countries are **taking charge of their lives and reproductive behavior.**
- women tend to have fewer children if they are educated.
- They have the **ability to control their own fertility**.
- individual women will play an important role in stabilizing populations, improving human health, reducing poverty and environmental degradation, **if they are empowered.**
- In most societies, they have fewer rights, educational and economic opportunities than men have.
- Women do almost all of the world's domestic work, child care and provide health care at free of cost.
- "For most women, the only holiday is when you are asleep."
- While women account for 66% of all hours worked, they receive only 10% of the world's income, own just 2% of the world's land.
- They also make up 70% of the world's poor and 66% of illiterate adults.





Trisha Shetty, worked to abolish 12 % tax on sanitary products.

6S Model for Women Empowerment



AN INITIATIVE BY INDIAN GOVT.

SHIKSHA = EDUCATION



- Lack of education is the root cause for women exploitation and negligence.
- About 245 million Indian women cannot read or write - world's largest number of unlettered women.
- Enrolment and retention of girls in education is poor -- average years of schooling for girls is only 1.2 years as against 3.5 years for boys

- Increases self confidence – become self-reliant and independent, make their own choices
- helps to builds a positive self-image and boosts self-confidence
- develops ability to think critically, understand the right and wrong
- to understand the Indian's constitutional and legislative provisions
- eradicates social evils - stand up against oppression
- Reduces inequalities and functions as a means of improving their status within the family ad also to change their position in the society.

SWASTHYA = HEALTH



- Improving women's health strengthens women's economic empowerment
- Women empowerment begins with guarantee of their health and safety
- Alarming concerns on maternal health care - Maternal mortality in India is the second highest in the world. Close to 125,000 women die due to pregnancy and pregnancy-related illnesses every year.
- Several health programs have been put forward by Government and several other NGO's, but still there exists a wide gap between those under protection and those not.
- Good health and education makes them capable to care for their children and others
- Studies show that nations that invest in women education and reproductive health, including family planning register slower population growth, higher productivity and faster economic growth.
- Women with proper health education are more likely to have fewer children, more likely to access health services and have control over health resources and less likely to suffer domestic violence.

SWAVLAMBAN = SELF RELIANCE



- This model leads to empowering women in rural areas and has gone a along way in developing society.
- Increases confidence and decision-making ability.
- Empower women from economically less privileged background to gain skills in various trades.
- Women participation in small scale activities in rural areas has made them self reliant.
- Self help groups are organized in several areas for assisting women in building micro entrepreneur skills to help them become economically self reliant.

SAMAJIK NYAY = JUSTICE

Women's access to **justice** is a critical part of this endeavor because when justice institutions are gender responsive, they support **women** to assert and claim their social, economic, cultural, political and civil rights. we are committed to promoting women's participation in the justice sector.

SAMVEDAN = SENSITIVITY

Being **sensitive** is, very simply put, being appreciative of others' feelings. In that context, gender **sensitivity** is about being considerate of the opposite gender's feelings. The reason this is **important** is because men and **women** think differently, and obviously, have varying perspectives.

SAMTA = EQUALITY

Gender equality prevents violence against women and girls. **Gender inequality** is a root cause of violence against women.

Urbanization

- **Enhancement of population in cities** by *natural increase or* by *immigration* from rural areas.
- It is the **process by which towns and cities are formed** and become larger as more people begin living and working in central areas.



Natural increase

High Birth Rate

Low death Rate

Better health care

Urban Sprawl

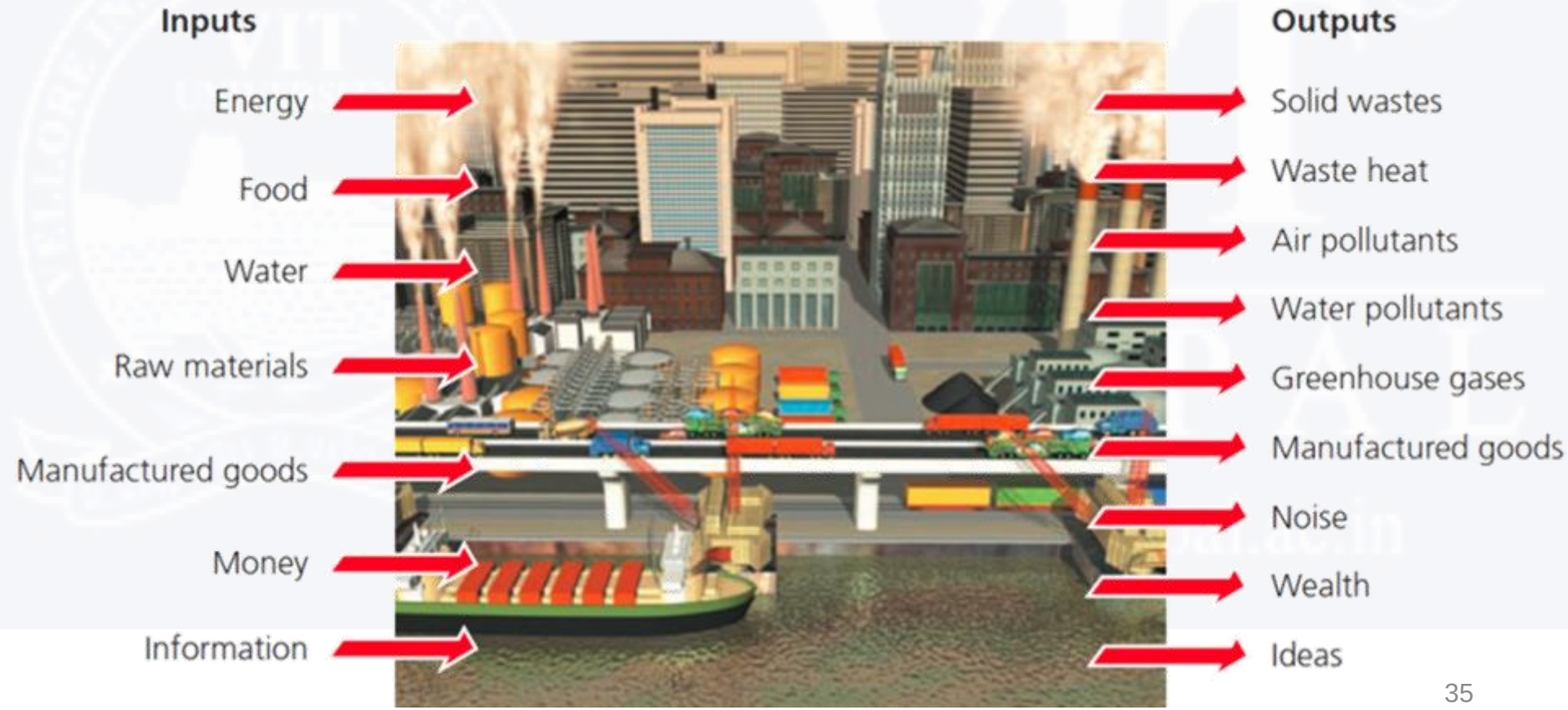
Immigration

jobs, food, housing, educational opportunities, Better health care, entertainment.

Most cities are unsustainable because of high levels of resource use, waste, pollution, and poverty.

Urbanization

- ✓ Urban populations occupy only about 3% of the earth's land area
- ✓ consume about 75% of its resources
- ✓ produce about 75% of the world's pollution and wastes.



Urbanization Has Disadvantages

- Most Cities Lack Vegetation.
- Many Cities Have Water Problems
- Concentrate Pollution and Health Problems
- Cities Have Excessive Noise
- Affect Local Climates
- Life Is a Desperate Struggle
- **Housing Problems**
- **Solid wastes and hazardous wastes**
- Transport problems
- Urban decay
- Social problems







Environmental Impacts

Impact on atmosphere and climate

- Creation of heat island.
- Change in air quality.
- Change in patterns and precipitation.

Impact on lithosphere and land resources

- Erosion and other changes in land quality.
- Pollution

Impact on hydrosphere and water resources

- Flow of Water into Streams.
- Degraded Water Quality.

Impact on biosphere

- Modification of Habitats.
- Destruction of Habitats.
- Creation of New Habitats.
- Loss of public space. The majority of roads are publicly owned and free of access.

Nature of housing problems

- Shortage of accommodation
- Substandard houses made of any materials
- Limited access to basic facilities such as clean water, electricity and sewage system
- Limited access to urban resources such as job opportunities, transport, education, medical and social services





Shortage of accommodation

- Growing squatter and slum population
- most immigrants tend to build themselves a shelter on land which does not belong to them.
- Or rent a single room in overcrowded and deteriorated slums.





Limited access to basic facilities

- Poor sanitation
- Poor living environment
- Outbreak of diseases







Limited access to urban resources

- Unemployment and underemployment
- family conflicts, mental distresses
- High crime rate
- Social unrest







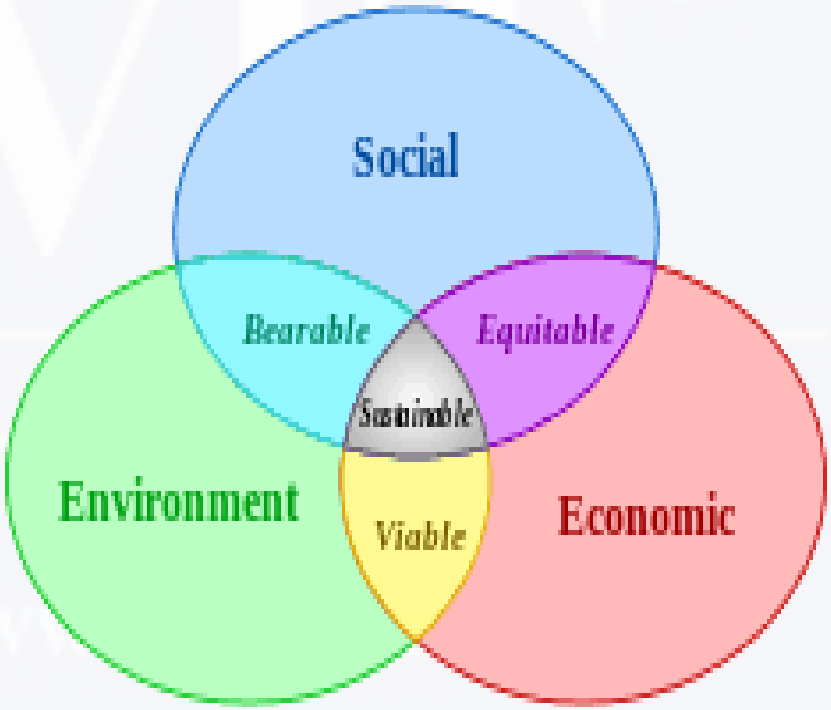






- **Smart Growth Can Promote Environmental Sustainability**
- **The Eco-City Concept: Cities for People, Not Cars**





Sustaining human societies

Limits and Regulations

Limit building permits

Draw urban growth boundaries

Create greenbelts around cities



Zoning

Promote mixed use of housing and small businesses

Concentrate development along mass transportation routes





Planning

Ecological land-use planning

Environmental impact analysis

Integrated regional planning



Protection

Preserve open space

Buy new open space

Prohibit certain types of development

Taxes

Tax land, not buildings

Tax land on value of actual use instead of on highest value as developed land

Tax Breaks

For owners agreeing not to allow certain types of development

For cleaning up and developing abandoned urban sites

Revitalization and New Growth

Revitalize existing towns and cities

Build well-planned new towns and villages within cities


Thank you!